

The growth in energy harvesting has accelerated with the start of Industry 4.0. The need for gathering data is growing at a rapid pace, which means more sensors are being installed even at the remotest of locations. Some application areas holding a great promise in the future are mentioned below.

Smart wearable technology.

Wearable technology is capable of tracking and monitoring different parameters of our body. This technology is in high demand in wearable applications, such as smart fitness, smart healthcare, smart military, etc. The smart wearable industry is growing at an unprecedented rate and there is a huge demand in consumer-grade wearable devices, such as fitness tracking devices like Fitbit and infotainment devices like Apple watch. The smart wearable has a great scope of

using piezo-electric transducers to generate energy and improve the battery life of the device. Mithras, a Switzerland based company, is developing TEGs compatible with wearable technology.

Self-powering medical implants.

Bio-medical is another field where a lot of research is going on to harness the energy inside our bodies to extend the lifetime of implants that require electrical energy, such as sensors to monitor body's parameters, pacemakers, and hearing aids, without changing the battery. Kinetic and heat energy can easily be harnessed. A joint mounted device based on generative braking can generate about 34W from the movement of knees and 20W from the ankles. Apart from thermal and kinetic energy harvesting, there are various other energy

harvesting options where research is going on; one such method is blood sugar oxidation.

Vibration and kinetic energy harvesting. Any movement by anyone or anything results in vibrations that can be harnessed to generate energy. The movement causes vibrations or low frequencies in the range of 1Hz to 30Hz. Piezoelectric transducers can be installed in sidewalks, bridges, roads, and public areas, such as hospitals, stadiums, etc. Companies like London based Pavegen use electromagnetic induction to generate energy from people walking on the sidewalks. **EFY**

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.

India's 1st Independent Survey



For details www.efy.in/elcina-imrb

